

- 7 The photograph shows different varieties of eggplant fruits.



(Source: © Leon Werdinger/Alamy Stock Photo)

The table shows the mass of eggplants imported by four countries in 2020.

Country	Mass imported in 2020/ tonnes
Austria	1 911
France	52 892
Italy	25 150
Slovenia	48

- (a) It is expected that the import of eggplants by France would increase by 1.6% per year.

Calculate the expected mass of eggplants imported by France in 2023.

Give your answer to the nearest whole number.

(2)

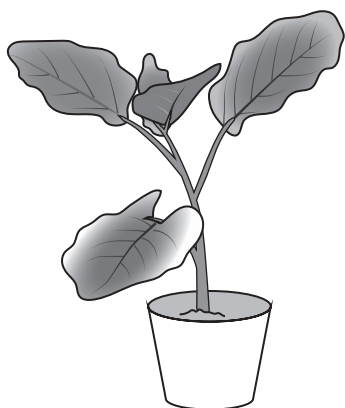
Answer tonnes

(b) *R. solanacearum* bacteria infect some varieties of eggplant.

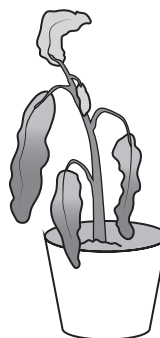
These bacteria reproduce in the xylem vessels and block them. This causes the disease bacterial wilt, which results in the death of the plant.

The extent of bacterial disease can be recorded as the disease index. The greater the extent of disease, the larger the value of the disease index.

The diagram shows a healthy plant and a wilted plant.



healthy plant



wilted plant

Scientists at a seed bank investigated the wilt percentage and disease index of some eggplant varieties.

Varieties	Wilt percentage (%)	Disease Index (DI)	Crop yield
A	95.8	45.8	high
B	66.7	61.3	medium
C	100.0	89.2	medium
D	33.3	9.2	low

The resistance of the plant variety was categorised based on its mean DI value as shown in the table.

Resistance category	DI value
Resistant	0–30
Moderately resistant	>30–40
Moderately susceptible	>40–50
Susceptible	>50

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- (4)

[illegible]

- (iii) Explain how scientists could breed new high-yielding varieties of eggplants which are resistant to *R. solanacearum* bacteria.

Use the information in the tables to support your answer.

(4)

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(Total for Question 7 = 11 marks)

